Q1:- Gradient Magnitude and Direction

Given Sobel operator values for a pixel

$G_x = 24$

$G_y = 18$

→ Gradient Magnitude (G):

$$G = \sqrt{(G_x)^2 + (G_y)^2}$$

$$= \sqrt{(24)^2 + (18)^2}$$

$$= \sqrt{900}$$

$$\boxed{G = 30}$$

→ Gradient direction (θ):

$$\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

$$\theta = \tan^{-1}\left(\frac{18}{24}\right)$$

$$= \tan^{-1}(0.75)$$

$$\theta = 36.87^\circ \approx 37^\circ$$

Q2: Non-Maximum Suppression (NMS)

Given that

Pixel	Magnitude
P1	25
P2	50
P3	90
P4	60
	30

P_1 (25) : Neighbor < P_2(50) : 0

P_2 (50) : Neighbor P3(90) : Suppressed(0)

P3(90) : Larger than P2(50) and P4(60) → kept(90)

P4 (60) : Neighbor. P3 (90) is larger → suppressed(0)

P5 (30) : Neighbor P4 (60) is larger → Suppressed(0)

Final NMS Magnitude:

* P1 : 0
* P_2 : 0
* P_3 : 90
* P4 : 0
* P5 : 0

Q3: Pixel Classification by Thresolding

Given that

High Threshold : 100

Low threshold : 50

Classification rules:

* Magnitude ≥ High threshold → strong edge
* Magnitude < low thresold → Non-edge
* Low threshold ≤ Magnitude < High threshold → weak edge

P1(135) : $\geq$ 100 : strong edge
P2(90) : < 100 : weak edge
P3(45) : < 50 : Non-edge
P4(120) : > 100 : strong edge
P5 (70) : < 100 : weak edge
P6 (20) : < 50 : Non-edge

Q4:- Hysteresis Thresholding magnitudes

Pixel	Magnitude	Initial classification
P1	130	strong edge
P2	95	weak edge
P3	40	Non-edge
P4	115	strong edge
P5	65	weak edge
P6	25	Non-edge
P7	105	strong edge

~~130 95 115~~

~~25~~

→ P2 is connected to P1

→ P5 is not connected to any strong edge (discard)

Final edge pixel hysteresis: P1, P2, P4, P7